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Cryptography

① Encrypt "CHASE YOUR DREAMS" using:

a. Caesar cipher

b. Shift cipher with key 7

c. ROT 13

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
15	16	17	18	19	20	21	22	23	24	25				
P	Q	R	S	T	U	V	W	X	Y	Z				

⇒ a) Caesar cipher: Formula: $C = (P+3) \bmod 26$

Plaintext: CHASE YOUR DREAMS

$$C \rightarrow (P+3) \bmod 26 = (2+3) \bmod 26 = 5 \bmod 26 = 5 = F$$

$$H \rightarrow (P+3) \bmod 26 = (7+3) \bmod 26 = 10 = K$$

$$A \rightarrow (P+3) \bmod 26 = (0+3) \bmod 26 = 3 = D$$

$$S \rightarrow (P+3) \bmod 26 = (18+3) \bmod 26 = 21 = V$$

$$E \rightarrow (P+3) \bmod 26 = (4+3) \bmod 26 = 7 = H$$

$$Y \rightarrow (P+3) \bmod 26 = (24+3) \bmod 26 = 1 = B$$

$$O \rightarrow (P+3) \bmod 26 = (14+3) \bmod 26 = 17 = R$$

$$U \rightarrow (P+3) \bmod 26 = (20+3) \bmod 26 = 23 = X$$

$$R \rightarrow (P+3) \bmod 26 = (17+3) \bmod 26 = 20 = U$$

$$D \rightarrow (P+3) \bmod 26 = (3+3) \bmod 26 = 6 = G$$

$$R \rightarrow (P+3) \bmod 26 = (17+3) \bmod 26 = 20 = U$$

$$E \rightarrow (P+3) \bmod 26 = (H+3) \bmod 26 = 7 = H$$

$$A \rightarrow (P+3) \bmod 26 = (O+3) \bmod 26 = 3 = D$$

$$M \rightarrow (P+3) \bmod 26 = (I+3) \bmod 26 = 15 = P$$

$$S \rightarrow (P+3) \bmod 26 = (U+3) \bmod 26 = 21 = V$$

Ciphertext: FKDVH BRXU GUHDPV

b) Shift cipher: Key: 7

$$\text{Formula: } C = E(P+7) \bmod 26$$

Plaintext: CHASE YOUR DREAMS

$$C \rightarrow (P+7) \bmod 26 = (2+7) \bmod 26 = 9 = J$$

$$H \rightarrow (P+7) \bmod 26 = (7+7) \bmod 26 = 14 = O$$

$$A \rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H$$

$$S \rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z$$

$$E \rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L$$

$$Y \rightarrow (P+7) \bmod 26 = (24+7) \bmod 26 = 5 = F$$

$$O \rightarrow (P+7) \bmod 26 = (14+7) \bmod 26 = 21 = V$$

$$U \rightarrow (P+7) \bmod 26 = (20+7) \bmod 26 = 1 = B$$

$$R \rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y$$

$$D \rightarrow (P+7) \bmod 26 = (3+7) \bmod 26 = 10 = K$$

$$R \rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y$$

$$E \rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L$$

$$A \rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H$$

$$M \rightarrow (P+7) \bmod 26 = (12+7) \bmod 26 = 19 = T$$

$$S \rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z$$

Ciphertext:

JOHXL FVBY KYLHTZ

C) ROT 13 : Formula: $C = E(P+13) \bmod 26$

Plaintext: CHASE YOUR DREAMS

$$C \rightarrow (P+13) \bmod 26 = (2+13) \bmod 26 = 15 = P$$

$$H \rightarrow (P+13) \bmod 26 = (7+13) \bmod 26 = 20 = U$$

$$A \rightarrow (P+13) \bmod 26 = (0+13) \bmod 26 = 13 = N$$

$$S \rightarrow (P+13) \bmod 26 = (18+13) \bmod 26 = 5 = F$$

$$E \rightarrow (P+13) \bmod 26 = (4+13) \bmod 26 = 17 = R$$

$$Y \rightarrow (P+13) \bmod 26 = (24+13) \bmod 26 = 11 = L$$

$$O \rightarrow (P+13) \bmod 26 = (14+13) \bmod 26 = 1 = B$$

$$U \rightarrow (P+13) \bmod 26 = (20+13) \bmod 26 = 7 = H$$

$$R \rightarrow (P+13) \bmod 26 = (17+13) \bmod 26 = 4 = E$$

$$D \rightarrow (P+13) \bmod 26 = (3+13) \bmod 26 = 16 = Q$$

$$R \rightarrow (P+13) \bmod 26 = (17+13) \bmod 26 = 4 = E$$

$$E \rightarrow (P+13) \bmod 26 = (4+13) \bmod 26 = 17 = R$$

$$A \rightarrow (P+13) \bmod 26 = (0+13) \bmod 26 = 13 = N$$

$$M \rightarrow (P+13) \bmod 26 = (12+13) \bmod 26 = 25 = Z$$

$$S \rightarrow (P+13) \bmod 26 = (18+13) \bmod 26 = 5 = F$$

Ciphertext: PUNFR LBHE QERNZF

② Decrypt:

a. "WRWD00B VHFUHW PHVVDJH" using Caesar Cipher

→ Caesar Cipher: Formula: $P = D(C-3) \bmod 26$

Ciphertext: WRWD00B VHFUHW PHVVDJH

$$W \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$$

$$R \rightarrow (C-3) \bmod 26 = (17-3) \bmod 26 = 14 = O$$

$$W \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$$

$$D \rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A$$

$$0 \rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L$$

$$0 \rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L$$

$$B \rightarrow (C-3) \bmod 26 = (1-3) \bmod 26 = 24 = Y$$

$$V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$$

$$H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$$

$$F \rightarrow (C-3) \bmod 26 = (5-3) \bmod 26 = 2 = C$$

$$U \rightarrow (C-3) \bmod 26 = (20-3) \bmod 26 = 17 = R$$

$$H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$$

$$W \rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T$$

$$P \rightarrow (C-3) \bmod 26 = (15-3) \bmod 26 = 12 = M$$

$$H \rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E$$

$$V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$$

$$V \rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S$$

$$D \rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A$$

$$J \rightarrow (C-3) \bmod 26 = (9-3) \bmod 26 = 6 = G$$

$$H \rightarrow (C-3) \bmod 26 = (1-3) \bmod 26 = 4 = E$$

Plaintext: TOTALLY SECRET MESSAGE

b. "SJB DTWP YNRJ" using shift cipher, 5

$\Rightarrow$ Shift cipher: Formula: $P = (C - 5) \bmod 26$

Ciphertext: SJB DTWP YNRJ

$$S \rightarrow (C-5) \bmod 26 = (18-5) \bmod 26 = 13 = N$$

$$J \rightarrow (C-5) \bmod 26 = (9-5) \bmod 26 = 4 = E$$

$$B \rightarrow (C-5) \bmod 26 = (1-5) \bmod 26 = 22 = W$$

$$D \rightarrow (C-5) \bmod 26 = (3-5) \bmod 26 = 24 = Y$$

$$T \rightarrow (C-5) \bmod 26 = (19-5) \bmod 26 = 14 = O$$

$$W \rightarrow (C-5) \bmod 26 = (22-5) \bmod 26 = 17 = R$$

$$P \rightarrow (C-5) \bmod 26 = (15-5) \bmod 26 = 10 = U$$

$$Y \rightarrow (C-5) \bmod 26 = (24-5) \bmod 26 = 19 = T$$

$$N \rightarrow (C-5) \bmod 26 = (13-5) \bmod 26 = 8 = I$$

$$R \rightarrow (C-5) \bmod 26 = (17-5) \bmod 26 = 12 = M$$

$$J \rightarrow (C-5) \bmod 26 = (9-5) \bmod 26 = 4 = E$$

Plaintext: NEW WORK TIME

c) "FRAQ ZBER PBSSRR" using ROT13

⇒ ROT 13: Formula: $P = (C + 13) \bmod 26$

Ciphertext: FRAQ ZBER PBSSRR

$$F \rightarrow (C + 13) \bmod 26 = (5 + 13) \bmod 26 = 18 = S$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$A \rightarrow (C + 13) \bmod 26 = (0 + 13) \bmod 26 = 13 = N$$

$$Q \rightarrow (C + 13) \bmod 26 = (16 + 13) \bmod 26 = 3 = D$$

$$Z \rightarrow (C + 13) \bmod 26 = (25 + 13) \bmod 26 = 12 = M$$

$$B \rightarrow (C + 13) \bmod 26 = (1 + 13) \bmod 26 = 14 = O$$

$$E \rightarrow (C + 13) \bmod 26 = (4 + 13) \bmod 26 = 17 = R$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$P \rightarrow (C + 13) \bmod 26 = (15 + 13) \bmod 26 = 2 = C$$

$$B \rightarrow (C + 13) \bmod 26 = (1 + 13) \bmod 26 = 14 = O$$

$$S \rightarrow (C + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = F$$

$$S \rightarrow (C + 13) \bmod 26 = (18 + 13) \bmod 26 = 5 = F$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

$$R \rightarrow (C + 13) \bmod 26 = (17 + 13) \bmod 26 = 4 = E$$

Plaintext: SEND MORE COFFEE